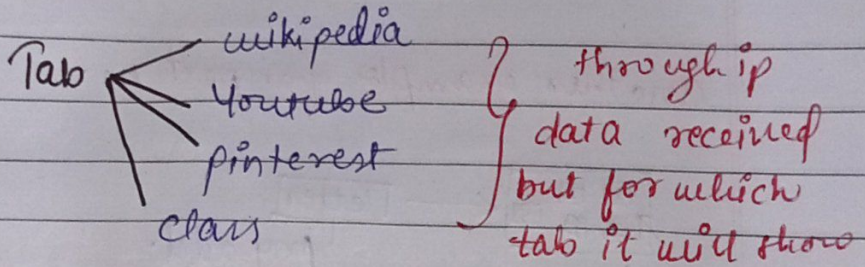
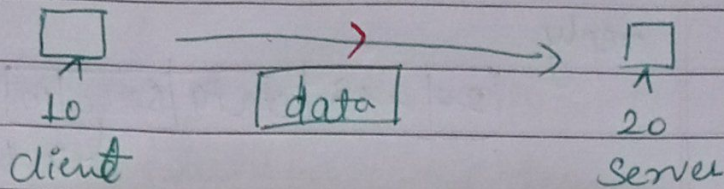


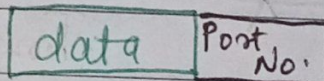
4. Transport layer

① - End to end delivery
(port to port)



Port Number

every website will have diff. port number



Port Number

→ 16 bits

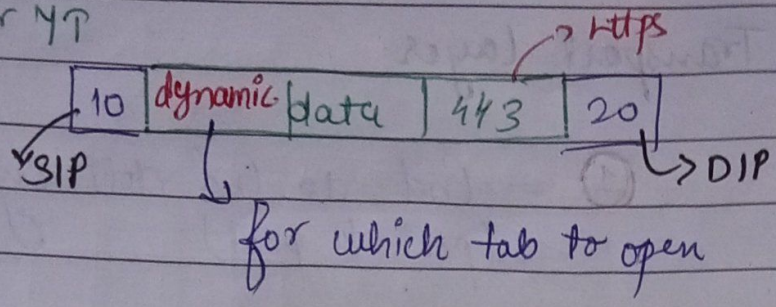
$2^{16} = 65536$ port no. can be generated
(0 - 65536)

Some ports are reserved

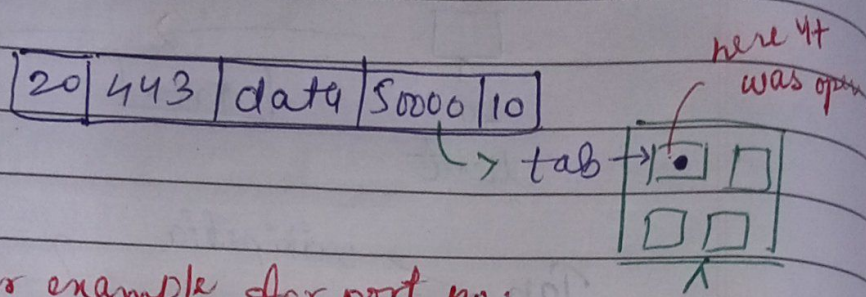
http: 40
https: 443
dns: 53

→ (0 - 1023) reserved
(1024 - 49151) - registered
(49152 - 65536) - dynamic/private
↳ for public

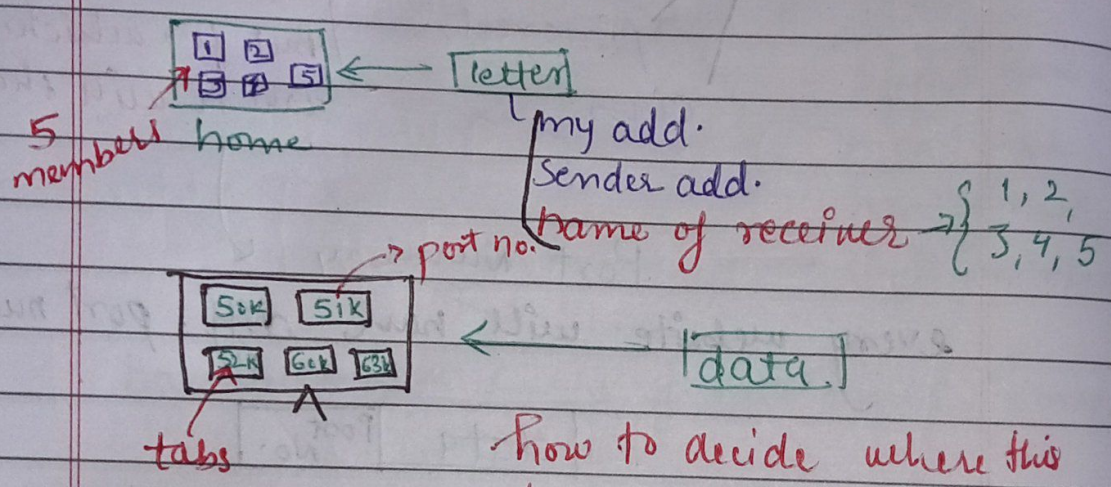
for YT



reply



another example for port no.



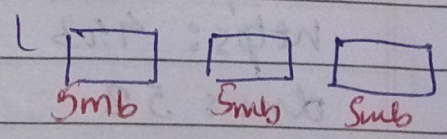
How to decide where this data will show

System provides port no. which is dynamic
↓
changes

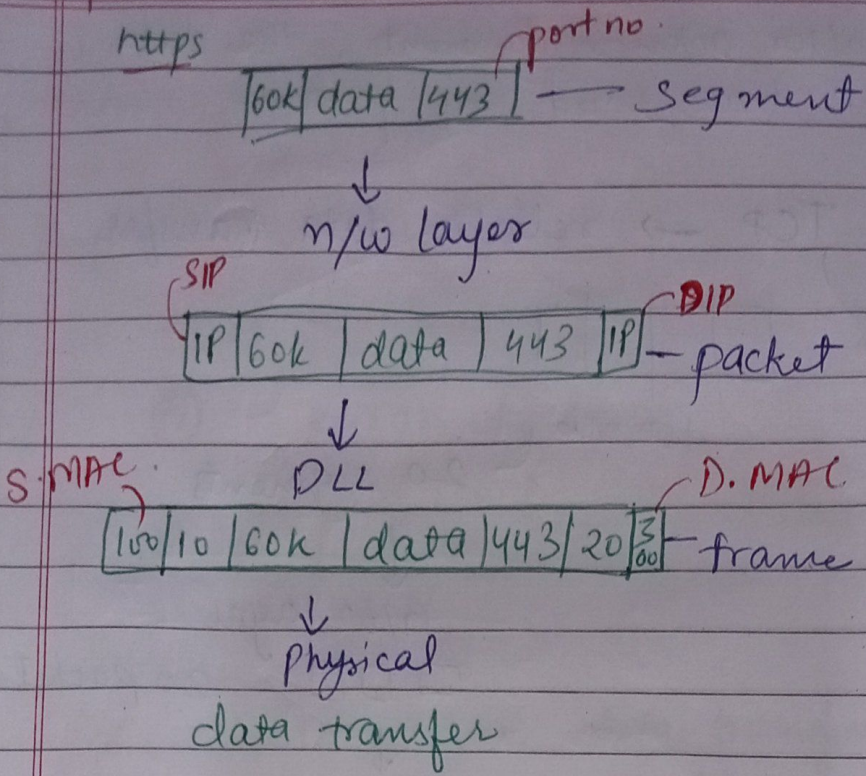
② - Segmentation

Transport layer

01010101 - Segments

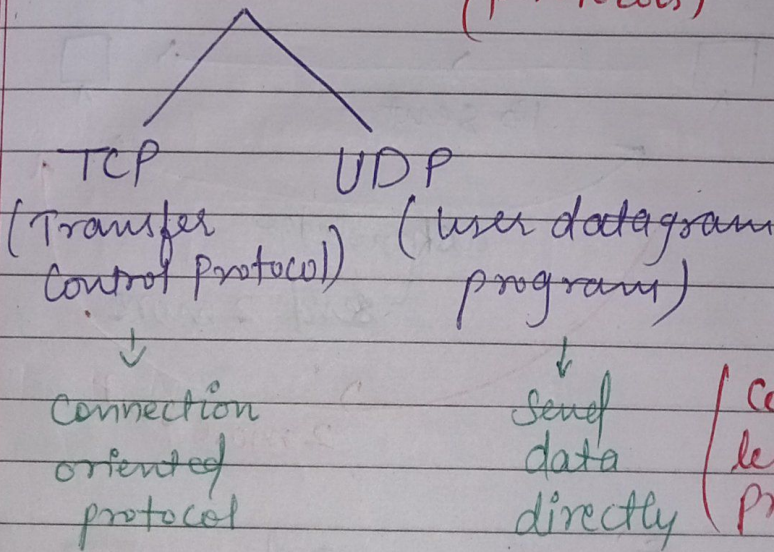


process →



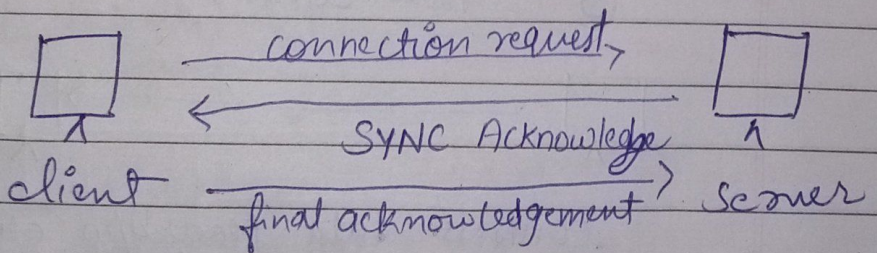
Transport Layer

(protocols)



at a time one is followed

Connection less protocol

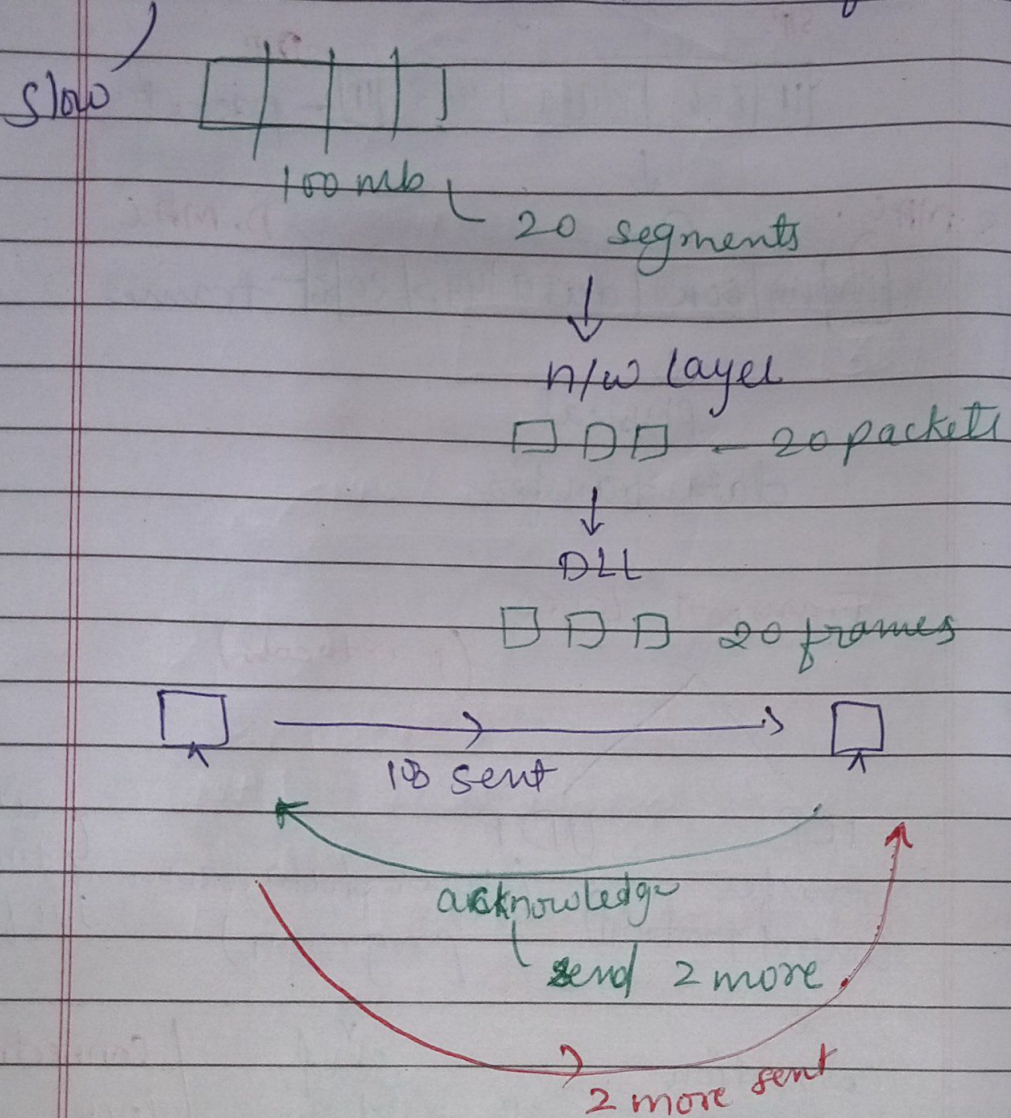


TCP

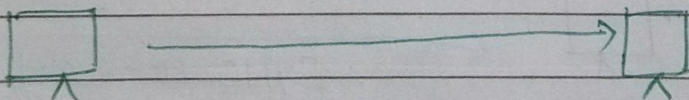
3 way handshake

after acknowledgement, the data is transferred by process.

TCP $\rightarrow$ reliable data transfer



③ — flow control



controls data flow b/w client & server.

(no router role)

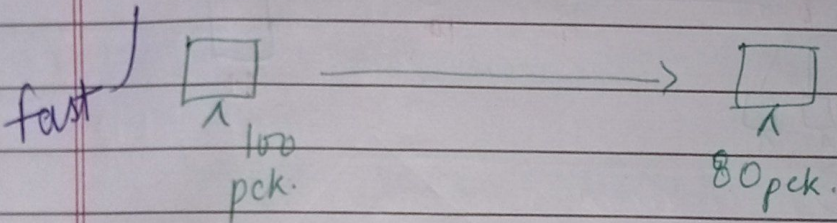
how do server knows how many packets are sent (to be sent)?

they discuss all while acknowledgement
(3 way hand-shake)

④ - error checking

20 to be sent but only client & server
& missing!

UDP → unreliable data transfer

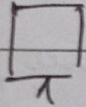


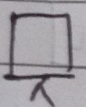
no flow control

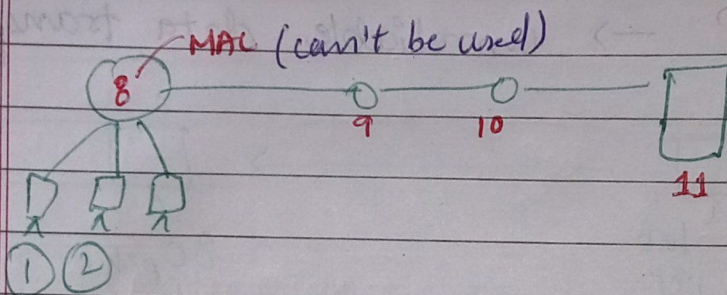
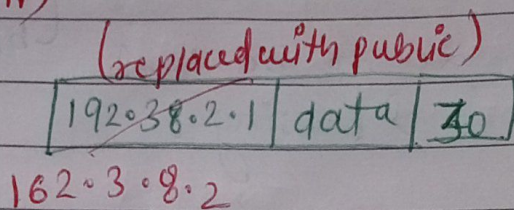
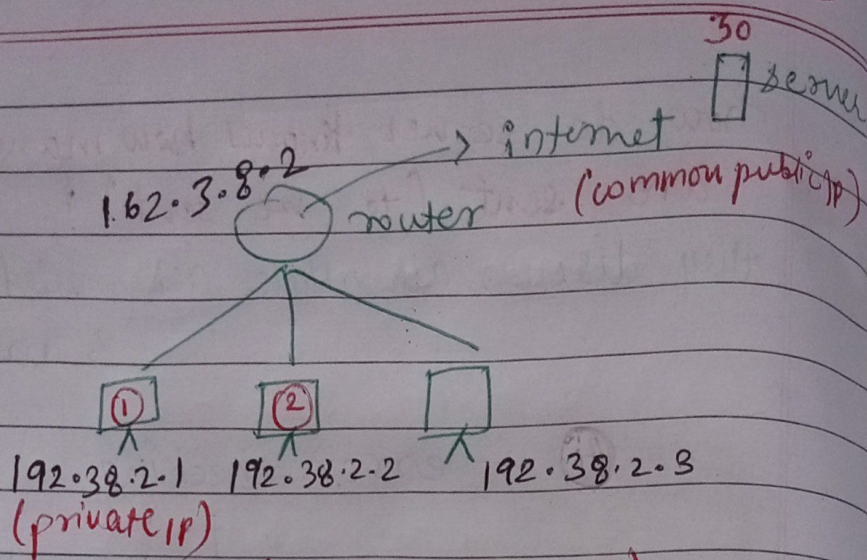
no way to check
20 are missing

↳ no connection made earlier

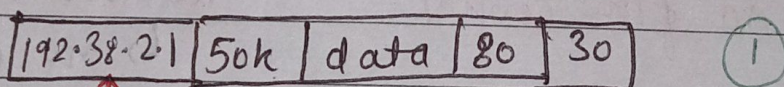
it depends on use case what protocol is best

TCP  wikipedia, (all data must be seen)
instagram,
email

UDP  livestream, (audio, video quality low but atleast data received without delay)
gaming(online)



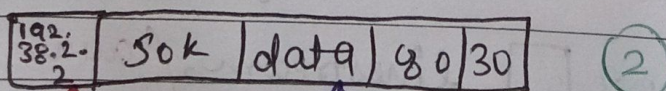
NAT: N/w address Translation



↑
router
will

change this to public IP

same time another system requests to same server with same port

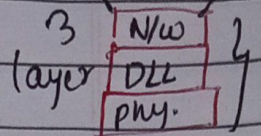


↑
change to
public IP

↑
data may
be different

How will server reply to them specifically

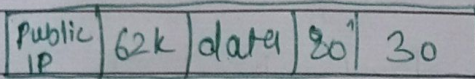
Router → NAT



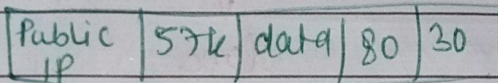
takes help from port no.

① 192.38.2.1 50k : 62k

↑
router will
give this private
port no.

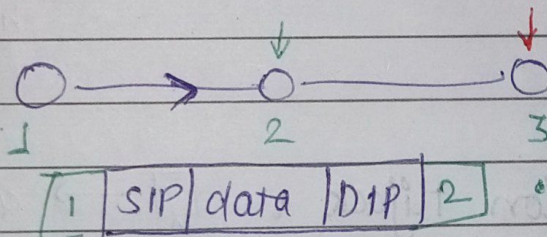


② 192.38.2.2 50k : 57k



Why new port no. ? MAC address why not ?

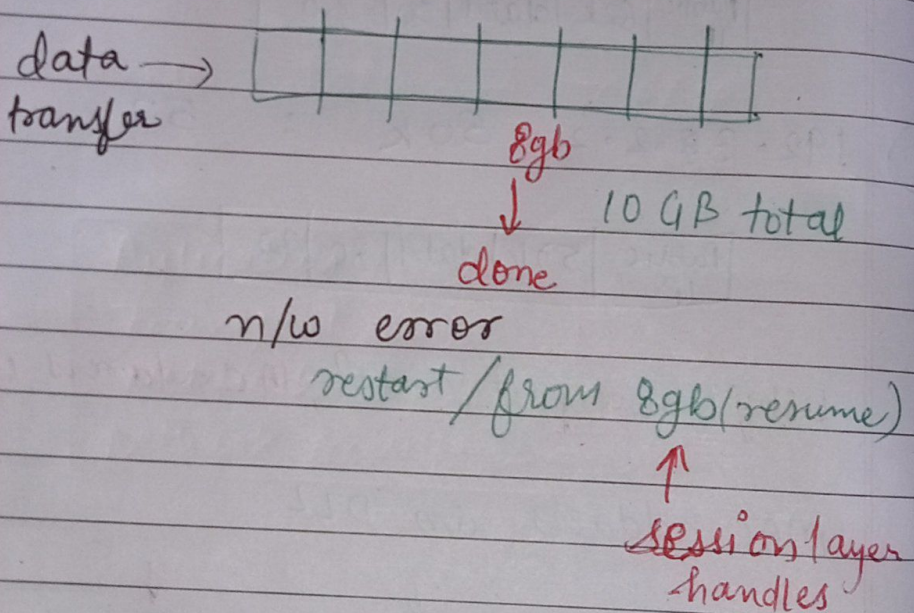
MAC address in DLL



MAC add. changes
while data transfer

5. Session layer

- ① - create session
- ② - authentication (id & password checks)
- ③ - authorisation
after login, what things you can access
- ④ - check point



Session Hijack :  → access
↓
cookies

⑥ Presentation layer

① - encrypt / decrypt

② - data compression

10mb → 6mb → data transfer
↑
presentation layer

③ - data translation

ASCII → UTF-16
□ □ understands

7. Application Layer

① - provide n/w interface

 YT UT → user sees
user ↔ Application layer
interaction

② - protocols

not secured ← http, ftp, smtp, https 3 services used
↓ security
file transfer (not used now)
mail transfer (gmail)

③ - network transparency

users only focus on use

↓
not focus on
internal working

(Application layer makes sure)

OSI

↳ theoretical model

implementation → TCP/IP model
(real)

Application layer	— App, presentation, session
Transport layer	
Network layer	
Network Access layer	— DLL + physical (in OSI)

Application → data received

Presentation → encrypt/decrypt

session → authenticate/authorise

Transport → segments □□□, port no.

Network → IP address

DLL → MAC address

Physical → data transfer

same process in receiver's end
(reverse order)